

PHISHING EMAIL DETECTION & AWARENESS REPORT

Future Interns

Cyber Security Internship

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INTRODUCTION

Phishing is a form of cyber attack in which attackers attempt to deceive users into revealing sensitive information such as passwords, banking details, One-Time Passwords (OTPs), or other confidential data.

Attackers commonly use fraudulent emails that appear to originate from trusted organizations, including banks, social media platforms, online services, and corporate institutions.

The primary objective of phishing attacks is to manipulate users through social engineering techniques rather than exploiting technical vulnerabilities.

Common phishing techniques include:

- Fake login pages designed to steal credentials
- Fraudulent emails impersonating trusted organizations
- Malicious links directing users to fake websites
- Urgent messages intended to create panic and force immediate action
- Requests for sensitive information such as passwords, OTPs, or financial details

Phishing attacks continue to be one of the most common cybersecurity threats because they target human behavior and decision-making rather than computer systems.

This report analyzes phishing email samples, identifies phishing indicators, classifies associated risks, and provides awareness guidelines to help users recognize and avoid phishing attacks.

OBJECTIVES

The objectives of this phishing email detection and awareness assessment are:

1. Analyze Email Samples

To examine email content and identify characteristics commonly associated with phishing attacks.

2. Identify Phishing Indicators

To detect warning signs such as suspicious sender addresses, urgent language, generic greetings, and malicious links.

3. Classify Email Risk Levels

To categorize emails as Safe, Suspicious, or Phishing based on the identified indicators and associated risks.

4. Understand Social Engineering Techniques

To study how attackers manipulate users through fear, urgency, trust, and deception.

5. Develop User Awareness

To educate users about recognizing phishing attempts and making informed security decisions.

6. Provide Prevention Guidelines

To recommend practical security measures that help users avoid becoming victims of phishing attacks.

7. Promote Cybersecurity Best Practices

To encourage safe email handling practices within organizations and among individual users.

PHISHING EMAIL SAMPLE

Introduction

The following email sample was analyzed to identify phishing indicators and assess the potential risk posed to users. The email contains several characteristics commonly associated with phishing attacks, including urgency tactics, suspicious links, and generic greetings.

```
|From: security@secure-account-verify.com  
Subject: Urgent: Your Account Will Be Locked  
  
Dear User,  
  
We noticed suspicious activity on your account.  
  
To avoid account suspension, please verify your details immediately.  
  
Verify Now:  
http://secure-account-verify.com  
  
Failure to verify within 24 hours will result in permanent account lock.  
  
Regards,  
Security Team
```

Initial Observation:

At first glance, the email appears to be a security notification. However, several suspicious elements are visible, including an unfamiliar sender domain, urgent language, a generic greeting, and a suspicious verification link. These characteristics indicate a potential phishing attempt and require further investigation.

PHISHING EMAIL ANALYSIS

Introduction :

The analyzed email was examined for common phishing indicators. Multiple red flags were identified, indicating that the email was designed to manipulate users into taking unsafe actions.

Analysis :

1. Suspicious Sender Domain

The email was sent from an unfamiliar domain that cannot be verified as a trusted organization.

2. Urgent Subject Line

The subject creates pressure by warning that the account will be locked.

3. Generic Greeting

The email uses "Dear User" instead of addressing the recipient by name.

4. Suspicious Link

The email contains an untrusted verification link that may redirect users to a fraudulent website.

5. Fear-Based Language

The message uses threats and urgency to encourage immediate action.

Analyst Conclusion

The email contains multiple phishing indicators and has been classified as a phishing attempt.

Classification : Phishing

Risk Level : High

LEGITIMATE EMAIL SAMPLE & ANALYSIS

From: [support@microsoft.com]
Subject: Password Reset Confirmation
Dear Vivek,
Your password was successfully changed on June 22, 2026.
If you did not make this change, please contact Microsoft Support through the official website.
Regards,
Microsoft Security Team

1. Trusted Sender Domain

The email originates from a recognized company domain (microsoft.com).

2. Personalized Greeting

The recipient is addressed by name.

3. Informational Tone

The email provides information without creating urgency or panic.

4. No Sensitive Information Request

The email does not ask for passwords, OTPs, or personal details.

5. Professional Communication

The message uses clear and professional language.

Classification : Safe

Risk Level : Low

RISK CLASSIFICATION

Risk Analysis :

The phishing email contains multiple indicators including a suspicious sender domain, generic greeting, urgency tactics, and an untrusted verification link.

These indicators increase the likelihood of credential theft, identity compromise, and unauthorized access to user accounts.

The legitimate email originates from a trusted domain, uses a personalized greeting, and does not request sensitive information, resulting in a low-risk classification.

Final Assessment :

The phishing email was classified as High Risk due to the presence of multiple phishing indicators.

The legitimate email was classified as Low Risk because it demonstrates characteristics of authentic and secure communication.

PHISHING PREVENTION GUIDELINES

1. Verify the Sender

Always check the sender's email address before responding to any email.

2. Avoid Clicking Suspicious Links

Do not click links from unknown or untrusted sources.

3. Check for Urgency Tactics

Be cautious of emails demanding immediate action or threatening consequences.

4. Look for Generic Greetings

Legitimate organizations usually address users by name.

5. Never Share Sensitive Information

Do not share passwords, OTPs, banking details, or personal information through email.

6. Verify Requests Independently

Contact the organization through official channels if an email appears suspicious.

7. Report Suspicious Emails

Inform the IT or Security Team when a phishing email is detected.

DO'S

1. Verify sender email addresses
2. Check links before clicking
3. Report suspicious emails
4. Use strong passwords and MFA
5. Confirm requests through official channels

DONT'S

1. Do not click unknown links
2. Do not download unexpected attachments
3. Do not share passwords or OTPs
4. Do not respond to suspicious emails
5. Do not trust urgency-based messages immediately

CONCLUSION

This report analyzed email samples to identify phishing indicators and evaluate associated security risks.

The phishing email sample contained multiple warning signs, including a suspicious sender domain, urgency tactics, generic greetings, and an untrusted verification link. Based on these findings, the email was classified as a phishing attempt with a high-risk level.

The legitimate email sample demonstrated characteristics of authentic communication, including a trusted sender domain, personalized greeting, professional language, and the absence of requests for sensitive information. Therefore, it was classified as safe with a low-risk level.

The assessment highlights the importance of user awareness in defending against phishing attacks. By verifying sender information, avoiding suspicious links, and following cybersecurity best practices, users can significantly reduce the risk of becoming victims of phishing attacks.

Security awareness remains one of the most effective defenses against social engineering and phishing threats.